Quantum 2 HW3

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Technion

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1 Simple Model for Quantization of the Electromagnetic Field

1. Use Maxwell's equations to write time development equations for $\mathcal{E}$ and $\mathcal{B}$.
2. Express $U(t)$ using $V, \mathcal{E}(t), \mathcal{B}(t), \mu_0, \varepsilon_0$. Use

$$\int_V \mathrm{d}\boldsymbol{r} \sin^2 kz = \int_V \mathrm{d}\boldsymbol{r} \cos^2 kz = \frac{V}{2}$$

3. We'll define $\omega = ck$ and

$$\chi(t) = \sqrt{\frac{\varepsilon_o V}{2\hbar\omega}}\mathcal{E}(t) \qquad \Pi(t) = \sqrt{\frac{V}{2\mu_0\hbar\omega}}\mathcal{B}(t)$$

show that the equations of motion for $\chi(t)$ and $\Pi(t)$ are identical in their form to those of X and P in subsection 1.

1.1 Time Development

The relevant Maxwell's equations for time development are

$$\nabla \times \boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t} \qquad \nabla \times \boldsymbol{B} = \mu_0 \left(\boldsymbol{J} + \varepsilon_0 \frac{\partial \boldsymbol{E}}{\partial t}\right) \tag{1}$$

and since no current or charges at all other than out waves were specified, we assume the current $\boldsymbol{J} = 0$. Plugging the given fields into the equations we find

$$k\mathcal{E} = \frac{\partial \mathcal{B}}{\partial t} \qquad k\mathcal{B} = -\mu_0\varepsilon_0 \frac{\partial \mathcal{E}}{\partial t} \tag{2}$$

we can take the derivative of the equations and plug them into the other ones to find 2nd order differential equations

$$\frac{\partial^2 \mathcal{E}}{\partial t^2} + \frac{k^2}{\mu_0\varepsilon_0}\mathcal{E} = 0 \qquad \frac{\partial^2 \mathcal{B}}{\partial t^2} + \frac{k^2}{\mu_0\varepsilon_0}\mathcal{B} = 0 \tag{3}$$

1.2 Energy in Time

$$U(t) = \int_V \mathrm{d}\boldsymbol{r} \left(\frac{\varepsilon_0}{2} \boldsymbol{E}^2 + \frac{1}{2\mu_0} \boldsymbol{B}^2 \right) \tag{4}$$

$$= \int_V \mathrm{d}\boldsymbol{r} \left(\frac{\varepsilon_0}{2} \mathcal{E}^2 \sin^2(kz) + \frac{1}{2\mu_0} \mathcal{B}^2 \cos^2(kz) \right) \tag{5}$$

$$= \frac{V}{2} \left(\frac{\varepsilon_0}{2} \mathcal{E}^2 + \frac{1}{2\mu_0} \mathcal{B}^2 \right) \tag{6}$$

1.3 Same Form

The equations of motion for $\mathcal{E}$ and $\mathcal{B}$ are 2nd order linear ODEs, which are invariant to multiplication by scalars. The transformation of parameters from $\mathcal{E}$ and $\mathcal{B}$ to χ and Π simply multiplies by a scalar, therefore the equations of motion will remain the same.

2 Quantization of the Field

1. Express $\hat{H}_F$ in terms of $\hat{a}$ and $\hat{a}^\dagger$ and the number operator $\hat{N} = \hat{a}^\dagger \hat{a}$.
2. Calculate the expectation values $\langle \hat{E}(\boldsymbol{r}) \rangle_n, \langle \hat{B}(\boldsymbol{r}) \rangle_n, \langle \hat{H}_F \rangle$ in state $|n\rangle$.

2.1 Field Hamiltonian

From the definitions for the photon creation and annihilation operators we can find expressions for χ and Π using the operators:

$$\hat{\chi} = \frac{1}{\sqrt{2}}\left(\hat{a} + \hat{a}^\dagger\right) \qquad \hat{\Pi} = \frac{1}{\sqrt{2}i}\left(\hat{a} - \hat{a}^\dagger\right) \tag{7}$$

therefore

$$\begin{aligned}
\hat{H}_F &= \frac{\hbar\omega}{2}\left(\hat{\chi}^2 + \hat{\Pi}^2\right) && (8)\\
&= \frac{\hbar\omega}{4}\left(\left(\hat{a} + \hat{a}^\dagger\right)^2 - \left(\hat{a} - \hat{a}^\dagger\right)\right) && (9)\\
&= \frac{\hbar\omega}{4}\left(\hat{a}^2 + \hat{a}\hat{a}^\dagger + \hat{a}^\dagger\hat{a} + \hat{a^\dagger}^2 - \hat{a}^2 + \hat{a}\hat{a}^\dagger + \hat{a}^\dagger\hat{a} - \hat{a^\dagger}^2\right) && (10)\\
&= \frac{\hbar\omega}{2}\left(\hat{a}\hat{a}^\dagger + \hat{N}\right) && (11)\\
&= \frac{\hbar\omega}{2}\left(\hat{a}^\dagger\hat{a} - \hat{a}^\dagger\hat{a} + \hat{a}\hat{a}^\dagger + \hat{N}\right) && (12)\\
&= \frac{\hbar\omega}{2}\left(2\hat{N} + [\hat{a}, \hat{a}^\dagger]\right) && (13)\\
&= \frac{\hbar\omega}{2}\left(2\hat{N} + 1\right) && (14)\\
&= \hbar\omega\left(\hat{N} + \frac{1}{2}\right) && (15)
\end{aligned}$$

2.2 Expectation Values

First we notice

$$\langle \hat{E}(\boldsymbol{r}) \rangle_n = \langle n| \hat{E} |n\rangle \tag{16}$$

which if we look at the definition of the creation and annihilation operators which define $\hat{E}$, the result of the above equation is always zero, therefore

$$\langle \hat{E}(\boldsymbol{r}) \rangle_n = 0 \tag{17}$$

and the same holds for $\langle \hat{B}(\boldsymbol{r}) \rangle_n$. For $\hat{H}_F$:

$$\langle n| \hat{H}_F |n\rangle = \hbar\omega \langle n| \left(\hat{N} + \frac{1}{2}\right) |n\rangle = \hbar\omega\left(n + \frac{1}{2}\right) \tag{18}$$

3 Coherent States of the Electromagnetic Field

The coherent states of the Electromagnetic field are defined as

$$|\alpha\rangle = e^{-\frac{|\alpha|^2}{2}} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle, \qquad \alpha \in \mathbb{C} \tag{19}$$

1. Show that the state $|\alpha\rangle$ is a normalized eigenstate of the annihilation operator a. Find the corresponding eigenvalue. Calculate the expectation value $\langle N \rangle_\alpha$ in this state.
2. Given $|\psi(t=0)\rangle = |\alpha\rangle$, show that
$$|\psi(t)\rangle = \exp\left(-i\frac{\omega t}{2}\right) |\alpha e^{-i\omega t}\rangle$$
3. Calculate the expectation values $\langle \boldsymbol{E}(\boldsymbol{r})\rangle_t$ and $\langle \boldsymbol{B}(\boldsymbol{r})\rangle_t$ in the state $|\psi(t)\rangle$. Assume α is real.
4. Show that $\langle \boldsymbol{E}(\boldsymbol{r})\rangle_t$ and $\langle \boldsymbol{B}(\boldsymbol{r})\rangle_t$ satisfy Maxwell's equations.
5. Calculate the classical energy of the field for $\boldsymbol{E}_{cl} = \langle \boldsymbol{E}(\boldsymbol{r})\rangle_t$ and $\boldsymbol{B}_{cl} = \langle \boldsymbol{B}(\boldsymbol{r})\rangle_t$ and compare with $\langle H_F \rangle_t$.
6. Assume $|\alpha| \gg 1$ and justify the statement that the states $|\alpha\rangle$ have a "classical" character.

3.1 Normalized Eigenstate

To show $|\alpha\rangle$ is an eigenstate of the annihilation operator we'll activate it on the state:

$$\begin{aligned}
a|\alpha\rangle &= e^{-\frac{|\alpha|^2}{2}} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} \cdot a|n\rangle & (20)\\
&= e^{-\frac{|\alpha|^2}{2}} \sum_{n=1}^{\infty} \frac{\alpha^n}{\sqrt{n!}} \sqrt{n}\,|n-1\rangle & (21)\\
&= e^{-\frac{|\alpha|^2}{2}} \sum_{n=1}^{\infty} \frac{\alpha^n}{\sqrt{(n-1)!}} |n-1\rangle & (22)\\
&= e^{-\frac{|\alpha|^2}{2}} \sum_{n=0}^{\infty} \frac{\alpha^{n+1}}{\sqrt{n!}} |n\rangle & (23)\\
&= \alpha e^{-\frac{|\alpha|^2}{2}} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} \cdot a|n\rangle & (24)\\
&= \alpha|\alpha\rangle & (25)
\end{aligned}$$

therefore $|\alpha\rangle$ is an eigenstate of the annihilation operator with eigenvalue α. In line 21 the sum begins from 1 because the ground state annihilates into a zero, so it can be removed from the sum. Next we check that the state is normalized:

$$\begin{aligned}
\langle\alpha|\alpha\rangle &= e^{-\frac{|\alpha|^2}{2}}\sum_{n=0}^{\infty}\frac{\alpha^n}{\sqrt{n!}}\cdot e^{-\frac{|\alpha^*|^2}{2}}\sum_{n=0}^{\infty}\frac{(\alpha*)^n}{\sqrt{n!}}\langle n|n\rangle &(26)\\
&= e^{-|\alpha|^2}\sum_{n=0}^{\infty}\frac{\alpha^{*n}\cdot\alpha^n}{n!} &(27)\\
&= e^{-|\alpha|^2}\sum_{n=0}^{\infty}\frac{|\alpha|^{2n}}{n!} &(28)\\
&= e^{-|\alpha|^2}\cdot e^{|\alpha|^2} &(29)\\
&= 1 &(30)
\end{aligned}$$

noting that in the general case it should be expanded as $\langle n|m\rangle$ but since it obviously becomes δ_{nm} I just skipped there directly. Next the expectation value of N is:

$$\langle N\rangle_\alpha = \langle\alpha|N|\alpha\rangle = \langle\alpha|a^\dagger a|\alpha\rangle = \alpha^*\alpha \tag{31}$$

3.2 Showing That

We know that $|\alpha\rangle$ is an eigenstate of the annihilation operator with eigenvalue α. As we know from question 2, the hamiltonian (which is time independent) of the EM field can be expressed using the creation and annihilation operators. This means the state $|\alpha\rangle$ is also an eigenstate of the EM field, which means we can use standard time evolution of eigenstates inserting the energy from question 2:

$$\begin{aligned}
|\psi(t)\rangle &= e^{-i\omega t\left(N+\frac{1}{2}\right)}|\alpha\rangle &(32)\\
&= e^{-\frac{|\alpha|^2}{2}}\sum_{n=0}^{\infty}e^{-i\omega t\left(n+\frac{1}{2}\right)}\frac{\alpha^n}{\sqrt{n!}}|n\rangle &(33)\\
&= e^{-\frac{|\alpha|^2-i\omega t}{2}}\sum_{n=0}^{\infty}\frac{\left(\alpha\cdot e^{-i\omega t}\right)^n}{\sqrt{n!}}|n\rangle &(34)\\
= \left[\beta\equiv\alpha\cdot e^{-i\omega t}\right] &= e^{-\frac{|\beta|^2-i\omega t}{2}}\sum_{n=0}^{\infty}\frac{\beta^n}{\sqrt{n!}}|n\rangle &(35)
\end{aligned}$$

so

$$|\psi(t)\rangle = e^{-\frac{i\omega t}{2}}|\beta\rangle = \exp\left(-i\frac{\omega t}{2}\right)|\alpha e^{-i\omega t}\rangle \tag{36}$$

3.3 Expectation Values

$$\begin{aligned}
\langle\boldsymbol{E}(\boldsymbol{r})\rangle_t &= \langle\psi(t)|\boldsymbol{E}|\psi(t)\rangle &(37)\\
&= \langle\alpha e^{i\omega t}|\boldsymbol{E}|\alpha e^{-i\omega t}\rangle &(38)\\
&= \varepsilon_x\sqrt{\frac{\hbar\omega}{\varepsilon_0 V}}\sin(kz)\langle\alpha e^{i\omega t}|\left(\alpha^\dagger+\alpha\right)|\alpha e^{-i\omega t}\rangle &(39)\\
&= \varepsilon_x\sqrt{\frac{\hbar\omega}{\varepsilon_0 V}}\sin(kz)\left(\alpha e^{-i\omega t}+\alpha e^{i\omega t}\right) &(40)\\
&= 2\alpha\varepsilon_x\sqrt{\frac{\hbar\omega}{\varepsilon_0 V}}\sin(kz)\cos(\omega t) &(41)
\end{aligned}$$

and

$$\langle \boldsymbol{B}(\boldsymbol{r})\rangle_t = \langle \alpha e^{i\omega t}|\boldsymbol{B}|\alpha e^{-i\omega t}\rangle \tag{42}$$

$$= i\varepsilon_y \sqrt{\frac{\mu_0 \hbar\omega}{V}} \cos(kz) \langle \alpha e^{i\omega t}| \left(a^\dagger - a\right) |\alpha e^{-i\omega t}\rangle \tag{43}$$

$$= i\varepsilon_y \sqrt{\frac{\mu_0 \hbar\omega}{V}} \cos(kz) \left(\alpha e^{i\omega t} - \alpha e^{-i\omega t}\right) \tag{44}$$

$$= -2\alpha\varepsilon_y \sqrt{\frac{\mu_0 \hbar\omega}{V}} \cos(kz) \sin(\omega t) \tag{45}$$

3.4 Satisfying Maxwell's Equations

For the first equation, on one hand

$$\nabla \times \langle E\rangle = \nabla \times \left(2\alpha\varepsilon_x \sqrt{\frac{\hbar\omega}{\varepsilon_0 V}} \sin(kz)\cos(\omega t)\right) \tag{46}$$

$$= 2\alpha k\varepsilon_y \sqrt{\frac{\hbar\omega}{\varepsilon_0 V}} \cos(kz)\cos(\omega t) \tag{47}$$

$$= [\omega = ck,\, c = \frac{1}{\sqrt{\mu_0\varepsilon_0}}] = 2\alpha ck \sqrt{\frac{\mu_0 \hbar ck}{V}} \cos(kz)\cos(ckt)\varepsilon_y \tag{48}$$

on the other hand

$$\frac{\partial\langle B\rangle}{\partial t} = -2\alpha\omega\varepsilon_y \sqrt{\frac{\mu_0 \hbar\omega}{V}} \cos(kz) \sin\cos\omega t \tag{49}$$

$$= -2\alpha ck \sqrt{\frac{\mu_0 \hbar ck}{V}} \cos(kz) \cos(ckt)\, \varepsilon_y \tag{50}$$

next

$$\nabla \times \langle B\rangle = \nabla \times \left(-2\alpha\varepsilon_y \sqrt{\frac{\mu_0 \hbar\omega}{V}} \cos(kz) \sin(\omega t)\right) \tag{51}$$

$$= -2\alpha k\varepsilon_x \sqrt{\frac{\mu_0 \hbar\omega}{V}} \sin(kz) \sin(\omega t) \tag{52}$$

$$= -2\alpha k \sqrt{\frac{\mu_0 \hbar ck}{V}} \sin(kz) \sin(ckt)\, \varepsilon_x \tag{53}$$

and

$$\mu_0\varepsilon_0 \frac{\partial\langle E\rangle}{\partial t} = -2\alpha\omega\varepsilon_x\mu_0 \sqrt{\frac{\varepsilon_0 \hbar\omega}{V}} \sin(kz)\sin(\omega t) \tag{54}$$

$$= -2\alpha k \sqrt{\frac{\mu_0 \hbar ck}{V}} \sin(kz)\sin(ckt)\varepsilon_x \tag{55}$$

3.5 Classical Energy

The classical energy of the EM field is

$$
\begin{aligned}
U &= \int_V \mathrm{d}\boldsymbol{r} \left(\frac{\varepsilon_0}{2} \langle E \rangle^2 + \frac{1}{2\mu_0} \langle B \rangle^2 \right) && (56) \\
&= \int_V \mathrm{d}\boldsymbol{r} \left(\frac{\varepsilon_0}{2} \left(2\alpha\varepsilon_x \sqrt{\frac{\hbar\omega}{\varepsilon_0 V}} \sin(kz)\cos(\omega t) \right)^2 + \frac{1}{2\mu_0} \left(-2\alpha\varepsilon_y \sqrt{\frac{\mu_0 \hbar\omega}{V}} \cos(kz)\sin(\omega t) \right)^2 \right) && (57) \\
&= 2\alpha^2 \frac{\hbar\omega}{V} \int_V \mathrm{d}\boldsymbol{r} \left(\sin^2(kz)\cos^2(\omega t) + \cos^2(kz)\sin^2(\omega t) \right) && (58) \\
&= \alpha^2 \hbar\omega && (59)
\end{aligned}
$$

In comparison, the energy from H_F is

$$
\langle H_F \rangle = \hbar\omega \left(\langle \psi | N | \psi \rangle + \frac{1}{2} \right) = \hbar\omega \left(\alpha^2 + \frac{1}{2} \right) \tag{60}
$$

So the classical energy is lower than the quantized energy.

3.6 Justification

If $|\alpha| \gg 1$ then the classical energy and the quantized energy are effectively the same, the $1/2$ term becoming too small to notice. In that case, the quantized behavior of the energy levels is effectively the same as the classical behavior.

4 Question 4

1. Show that the set of states A is an eigenbasis of $H_0 = H_A + H_F$. Find the eigenvalues.
2. Draw qualitatively on the energy axis the location of the five first eigenenergies of H_0. Show that aside from the ground state, H_0's eigenstates are bunched in pairs of degenerate states.
3. The coupling between the field and the atom is a dipole coupling and is given by

$$W = \gamma \left(a\sigma_+ + a^\dagger \sigma_-\right)$$

where $\gamma = -d \sin(kz_0) \sqrt{\hbar\omega/\varepsilon_0 V}$ where d is the dipole moment, determined experimentally.

   (a) Write the action of $\hat{W}$ on the states $|g, n\rangle$ and $|e, n\rangle$

   (b) Which physical processes to the operators $a\sigma_+$ and $a^\dagger\sigma_-$ describe?

4.1 Eigenbasis

I'll assume the meaning of the question is to prove the states in set A are eigenstates of the hamiltonian, not to prove they span the energy space.

$$\begin{aligned} H_0 |g, n\rangle &= \left(\frac{\hbar\omega_A}{2}\sigma_z + \frac{\hbar\omega}{2}\left(\chi^2 + \Pi^2\right)\right)|g, n\rangle && (61) \\ &= -\frac{\hbar\omega_A}{2} + \hbar\omega\left(n + \frac{1}{2}\right)|g, n\rangle && (62) \end{aligned}$$

and

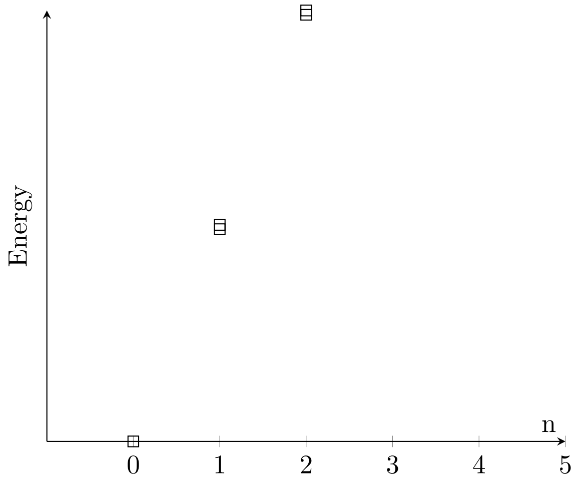
$$\begin{aligned} H_0 |e, n\rangle &= \left(\frac{\hbar\omega_A}{2}\sigma_z + \frac{\hbar\omega}{2}\left(\chi^2 + \Pi^2\right)\right)|e, n\rangle && (63) \\ &= \frac{\hbar\omega_A}{2} + \hbar\omega\left(n + \frac{1}{2}\right)|e, n\rangle && (64) \end{aligned}$$

4.2 Drawing

Since $\omega_A = \omega$ the energies are now

$$H_0 |g, n\rangle = \hbar\omega n |g, n\rangle \qquad H_0 |e, n\rangle = \hbar\omega (n + 1) |e, n\rangle \tag{65}$$

we can see that the ground state energy is 0, then every energy level has is degenerate with 2 options for achieving that energy.



4.3 Coupling

4.3.1 Action of W

$$\begin{aligned}
W |g, n\rangle &= \gamma \left(a\sigma_+ + a^\dagger \sigma_- \right) |g, n\rangle &(66)\\
&= \gamma a\sigma_+ |g, n\rangle &(67)\\
&= \gamma \sqrt{n} |e, n - 1\rangle &(68)\\
W |e, n\rangle &= \gamma \left(a\sigma_+ + a^\dagger \sigma_- \right) |e, n\rangle &(69)\\
&= \gamma a^\dagger \sigma_- |g, n\rangle &(70)\\
&= \gamma \sqrt{n + 1} |g, n + 1\rangle &(71)
\end{aligned}$$

and of course the $W |g, 0\rangle = 0$.

4.3.2 Physical Process

$a\sigma_+$ desribes a photon being absorbed by the atom raising it to the excited state. $a^\dagger \sigma_-$ describes the opposite.

5 Rabi Oscillations

1. Find the state of the system $|\psi(t)\rangle$ at a time $t > 0$.
2. What is the probability $P_g(T)$ to find the atom in the ground state $|g\rangle$ at time T, when the atom leaves the resonator? Show that $P_g(T)$ is a periodic function.
3. For Rubidium atoms $d = 1.1 \times 10^{-26}\,\mathrm{C\,m}$ and $\frac{\omega}{2\pi} = 5.0 \times 10^{10}\,\mathrm{Hz}$ and the experiment is done in a resonator of volume $V = 1.87 \times 10^{-6}\,\mathrm{m}^3$. The graph of $P_g(T)$ and of the real part of the fourier transform $J(\nu) = \int_0^\infty \mathrm{d}T P_g(T)\cos(2\pi\nu T)$ are presented. Compare theory to experiment.

5.1 State at Time

First we need to check if the hamiltonian $H = H_0 + W$ can even be diagonalized and for that we check if H_0 and W share an eigenbasis. To check that we look at their commutator

$$
\begin{aligned}
[H_0, W] &= \left[\left(\frac{\hbar\omega}{2}\sigma_z + \hbar\omega\left(N + \frac{1}{2}\right)\right), \gamma\left(a\sigma_+ + a^\dagger\sigma_-\right)\right] && (72)\\
&= \frac{\hbar\omega\gamma}{2}\left(\left[\sigma_z, a\sigma_+ + a^\dagger\sigma_-\right] + 2\left[N, a\sigma_+ + a^\dagger\sigma_-\right]\right) && (73)\\
&= \hbar\omega\gamma\left(a\sigma_+ - a^\dagger\sigma_- - a\sigma_+ + a^\dagger\sigma_-\right) && (74)\\
&= 0 && (75)
\end{aligned}
$$

So H can be diagonalized by diagonalizing H_0 and W together. We'll try to express the eigenstates of W using the eigenstates of H_0. Since W moves excited states to ground states and vice versa while removing or adding a photon respectively I'll guess normalized vectors of the from

$$
|\psi_{\pm,n}\rangle = \frac{1}{\sqrt{2}}\left(|e,n\rangle \pm |g,n+1\rangle\right) \tag{76}
$$

since they represent superpositions of states with an excited atom and n photons and ground state atoms with n+1 photons. We'll check

$$
\begin{aligned}
W|\psi_{+,n}\rangle &= W\frac{1}{\sqrt{2}}\left(|e,n\rangle + |g,n+1\rangle\right) && (77)\\
&= \gamma\left(a\sigma_+ + a^\dagger\sigma_-\right)\frac{1}{\sqrt{2}}\left(|e,n\rangle + |g,n+1\rangle\right) && (78)\\
&= \frac{\gamma}{\sqrt{2}}\left(a^\dagger|g,n\rangle + a|e,n+1\rangle\right) && (79)\\
&= \frac{\gamma}{\sqrt{2}}\sqrt{n+1}\left(|g,n+1\rangle + |e,n\rangle\right) && (80)\\
&= \gamma\sqrt{n+1}\,|\psi_{+,n}\rangle && (81)
\end{aligned}
$$

and

$$W\left|\psi_{-,n}\right\rangle = W\frac{1}{\sqrt{2}}\left(|e,n\rangle - |g,n+1\rangle\right) \tag{82}$$
$$= \gamma\left(a\sigma_+ + a^\dagger\sigma_-\right)\frac{1}{\sqrt{2}}\left(|e,n\rangle - |g,n+1\rangle\right) \tag{83}$$
$$= \frac{\gamma}{\sqrt{2}}\left(a^\dagger|g,n\rangle - a|e,n+1\rangle\right) \tag{84}$$
$$= \frac{\gamma}{\sqrt{2}}\sqrt{n+1}\left(|g,n+1\rangle - |e,n\rangle\right) \tag{85}$$
$$= -\gamma\sqrt{n+1}\left|\psi_{-,n}\right\rangle \tag{86}$$

So those are indeed eigenstates of W too, therefore these are also eigenstates of the full hamiltonian. The energies of these eigenstates are

$$E_{\pm,n} = \hbar\omega\left(n+1\right) \pm \gamma \tag{87}$$

We can now express the initial state as a combination of these

$$|\psi(0)\rangle = |e,0\rangle = \frac{1}{\sqrt{2}}\left(|\psi_{+,0}\rangle + |\psi_{-,0}\rangle\right) \tag{88}$$

and use the time independent propagator

$$|\psi(t)\rangle = \frac{1}{\sqrt{2}}U(t)\left(|\psi_{+,0}\rangle + |\psi_{-,0}\rangle\right) \tag{89}$$
$$= \frac{1}{\sqrt{2}}e^{-i\left(\omega+\frac{\gamma}{\hbar}\right)t}|\psi_{+,0}\rangle + \frac{1}{\sqrt{2}}e^{-i\left(\omega-\frac{\gamma}{\hbar}\right)t}|\psi_{-,0}\rangle \tag{90}$$
$$= \frac{1}{\sqrt{2}}e^{-i\omega t}\left(e^{-\frac{i\gamma}{\hbar}t}|\psi_{+,0}\rangle + e^{\frac{i\gamma}{\hbar}t}|\psi_{-,0}\rangle\right) \tag{91}$$
$$= \frac{1}{2}e^{-i\omega t}\left(e^{-\frac{i\gamma}{\hbar}t}\left(|e,0\rangle + |g,1\rangle\right) + e^{\frac{i\gamma}{\hbar}t}\left(|e,0\rangle - |g,1\rangle\right)\right) \tag{92}$$
$$= \frac{1}{2}e^{-i\omega t}\left(\left(e^{-\frac{i\gamma}{\hbar}t} + e^{\frac{i\gamma}{\hbar}t}\right)|e,0\rangle + \left(e^{-\frac{i\gamma}{\hbar}t} - e^{\frac{i\gamma}{\hbar}t}\right)|g,1\rangle\right) \tag{93}$$
$$= e^{-i\omega t}\left(\cos\left(\frac{\gamma t}{\hbar}\right)|e,0\rangle - i\sin\left(\frac{\gamma t}{\hbar}\right)|g,1\rangle\right) \tag{94}$$

5.2 Probability of Ground State

$$P_g(T) = \left|\langle g,1|\psi(T)\rangle\right|^2 = \sin^2\left(\frac{\gamma T}{\hbar}\right) \tag{95}$$

which is periodic.

5.3 Rubidium

We found the probability to be periodic and since it's a sine square the period is twice the expression in the sine So

$$\nu = f/2\pi = \frac{\gamma}{\pi\hbar} \tag{96}$$

$$= \frac{d\sin(kz_0)\sqrt{\hbar\omega/\varepsilon_0 V}}{\pi\hbar} \tag{97}$$

$$= 1.1\times10^{-26}\,\mathrm{C\,m}\cdot\sqrt{\frac{2\cdot 5.0\times10^{10}\,\mathrm{Hz}}{\pi\cdot 1.05\times10^{-34}\,\mathrm{J\,s}\cdot 8.85\times10^{-12}\,\mathrm{F\,m^{-1}}\cdot 1.87\times10^{-6}\,\mathrm{m^3}}} \tag{98}$$

$$= 47\,\mathrm{kHz} \tag{99}$$

which fits the peak in the frequency graph nicely.

6 Interactions Between the Atom and Coherent States

The system is prepared in the excited state $|e\rangle$ and the field in the coherent state $|\alpha\rangle$.

1. Calculate the probability $P_g(T,n)$ to find the atom in the ground state $|g\rangle$ and the field in the state $|n+1\rangle$ at time T. What is the probability of the state $|g,0\rangle$?
2. Find the probability $P_g(T)$ to find the atom in the ground state regardless of the field's state.
3. Experimental measurements are shown. The resonator is the same but the field was prepared in the coherent state. The most influential frequencies are

$$\nu_0 = 47\,\text{kHz} \qquad \nu_1 = 65\,\text{kHz} \qquad \nu_2 = 81\,\text{kHz} \tag{100}$$

   (a) Do the ratios $\frac{\nu_1}{\nu_0}$ and $\frac{\nu_2}{\nu_0}$ have the value you expected?

   (b) Use the values of $J(\nu_0)$ and $J(\nu_1)$ to estimate the number of photons $|\alpha|^2$ in the resonator.

6.1 Ground Probability

The probability is

$$P_g(T) = |\langle g, n+1|U(t)|e,\alpha\rangle|^2 \tag{101}$$

side calculation:

$$\begin{aligned}
\langle g, n+1|\, U(t) &= \frac{1}{\sqrt{2}}\left(\langle\psi_{+,n}| - \langle\psi_{-,n}|\right) U(t) && (102)\\
&= \frac{1}{\sqrt{2}} e^{i\omega T}\left(e^{i\frac{\gamma}{\hbar}\sqrt{n+1}T}\langle\psi_{+,n}| - e^{-i\frac{\gamma}{\hbar}\sqrt{n+1}T}\langle\psi_{-,n}|\right) && (103)\\
&= e^{i\omega T}\left(i\sin\left(\frac{\gamma}{\hbar}\sqrt{n+1}T\right)\langle e,n| + \cos\left(\frac{\gamma}{\hbar}\sqrt{n+1}T\right)\langle g,n+1|\right) && (104)
\end{aligned}$$

inserting back into Equation 101

$$\begin{aligned}
P_g(T) &= \left|\left(i\sin\left(\frac{\gamma}{\hbar}\sqrt{n+1}T\right)\langle e,n| + \cos\left(\frac{\gamma}{\hbar}\sqrt{n+1}T\right)\langle g,n+1|\right)|e,\alpha\rangle\right|^2 && (105)\\
&= \sin^2\left(\frac{\gamma}{\hbar}\sqrt{n+1}T\right)|\langle n|\alpha\rangle|^2 && (106)\\
&= \sin^2\left(\frac{\gamma}{\hbar}\sqrt{n+1}T\right)\left|e^{-\frac{|\alpha|^2}{2}}\sum_{m=0}^{\infty}\frac{\alpha^m}{\sqrt{m!}}\langle n|m\rangle\right|^2 && (107)\\
&= \sin^2\left(\frac{\gamma}{\hbar}\sqrt{n+1}T\right)e^{-|\alpha|^2}\frac{\alpha^{2n}}{n!} && (108)
\end{aligned}$$

The probability of finding the system in the state $|g,0\rangle$ is 0, because that would require the system to both give off energy to come down to the ground state but also for no photon to carry that energy away, which is not physical. Mathematically, there is no state $|e,\alpha\rangle$ with which it is paired through the interactions of the question so there is no way to reach it through the operation we defined from an excited state.

6.2 No Field Dependence

This is simply a sum over all field states for the previous part's probability

$$P_g(T) = \sum_{n=0}^{\infty} \left(\sin^2 \left(\frac{\gamma}{\hbar} \sqrt{n+1} T \right) e^{-|\alpha|^2} \frac{\alpha^{2n}}{n!} \right) \tag{109}$$

6.3 Coherent Experiment

6.3.1 Ratio values

The ratios are

$$\frac{\nu_1}{\nu_0} = \frac{65}{47} =\sim 1.38 \qquad\qquad \frac{\nu_2}{\nu_0} = \frac{81}{47} =\sim 1.72 \tag{110}$$

while the theoretical ratios are

$$\frac{\nu_1}{\nu_0} = \frac{\sqrt{2}}{\sqrt{1}} =\sim 1.4 \qquad\qquad \frac{\nu_2}{\nu_0} = \frac{\sqrt{3}}{\sqrt{1}} =\sim 1.73 \tag{111}$$

which are indeed quite similar.

6.3.2 Number of Photons

Since J is a fourier transform of P, they are both proportional to the same constants. Therefore

$$\frac{J(\nu_1)}{J(\nu_0)} = \frac{P(n=1)}{P(n=0)} = |\alpha|^2 \tag{112}$$

which from the graph looks like it is

$$|\alpha|^2 \approx \frac{4}{4.5} =\sim 0.88 \tag{113}$$